

Saber Ayatollahi, CFA, MSc

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SUMMARY

Senior Associate, Total Portfolio Rebalancing & Modelling (12-month Contract) candidate with ~7 years of quantitative portfolio research across multi-asset institutional portfolios. Hands-on VaR/CVaR portfolio optimization, risk decomposition, and near-optimal frontier analysis (Ortec GLASS) alongside Monte Carlo economic scenario-generator development; led the design of a Python-based multi-asset cash-flow projection engine at Moody's Analytics. Holds delegated sign-off on valuation and sensitivity outputs for institutional portfolios of \$5-25bn per engagement. Presents methodology and research findings to senior investment stakeholders. CFA; dual MSc (Financial Modelling, Chemical Engineering).

CORE SKILLS

Quantitative Research & Modelling • Portfolio Construction & Asset Allocation (SAA / TAA) • Portfolio Optimization - VaR & CVaR • Risk Decomposition & Attribution • Scenario Analysis & Stress Testing • Stochastic Modelling & Monte Carlo Simulation • Multi-Asset Portfolio Analytics • Python (pandas, NumPy, SciPy) • SQL • R • MATLAB • Model Development, Implementation & Testing

EXPERIENCE

Moody's Analytics

Asst. Director, Modelling Services

Toronto, May 2023 - Present

Multi-Asset Portfolio Modelling & Analytics

- Holds delegated sign-off authority on valuation, sensitivity, and portfolio-level analytics delivered to multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions prior to production release.
- Validated derivatives pricing outputs across rates, FX, and inflation, confirming consistency of sensitivities and scenario impacts at portfolio-level aggregates.
- Reviewed aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream allocation and capital processes.
- Oversaw duration and rate-sensitivity analysis under parallel and non-parallel yield-curve shocks across multi-asset institutional books.

Quantitative Model Development, Implementation & Testing

- Led design and implementation of an enterprise multi-asset cash-flow projection engine supporting base, stress, and reverse-stress scenarios across multi-year horizons.
- Re-engineered manual spreadsheet workflows into scalable Python (pandas, NumPy, SciPy) analytics pipelines with embedded logging, validation, and auditability controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection, reducing development cycle time an estimated 30-40% on comparable modules.

Methodology Review & Research Communication

- Contributes to methodology review, documentation, and benchmarking standards through Moody's model governance committee, supporting model-performance assessment for client-delivered analytics.
- Escalates outputs lacking mathematical or economic defensibility and acts as escalation point for client-critical analytical issues with clients' Heads of Risk and investment stakeholders.
- Prepared analytical summaries for senior stakeholders covering scenario impacts and portfolio sensitivities, translating quantitative output into investment-committee-ready narratives.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe DB (risk & ALM analytics) and PFaroe PM (portfolio management) platforms; validated model outputs post-deployment.
- Translated investment-team requirements into scoped development requests for Product Owners, contributing to the migration of all assigned accounts from the legacy Calypso platform.

Ernst & Young (EY)

Senior Consultant, Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning methodology requirements with system implementation and reporting frameworks.
- Coordinated across actuarial, finance, IT, risk, and program-management teams on milestone delivery, governance documentation, and implementation materials.

Ortec Finance Canada

Consultant, ALM & Risk

Toronto, Feb 2019 - Sep 2021

- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition, contribution-to-risk budgeting, and near-optimal frontier analysis to test the robustness of allocation recommendations.
- Built and interpreted stochastic economic scenario generators (Monte Carlo) assessing interest-rate risk and funding volatility under base-case and stressed environments.
- Advised institutional clients on Strategic Asset Allocation (SAA) and helped navigate Tactical Asset Allocation (TAA) risk through stress testing and sensitivity analysis.
- Led the ALM study for a three-plan university pension merger on a team of three, developing and validating the model capturing all Funding Policy dynamics and evaluating duration, currency, inflation, and leverage overlay strategies.
- Modelled actuarial liabilities alongside all major investment-product types and economic variables; performed cash-flow, liquidity, and hedging analysis across interest-rate, inflation, and currency risk.
- Presented study findings and allocation recommendations to investment committees at Canadian pension-fund clients on-site.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016